

MP307 Practical 2019/2020: Discrete Population Models

This practical uses the Maple worksheet `discrete.mw` that may be downloaded from the **MP307 Blackboard** pages.

Notice

Solutions to **all** questions marked with (*) has to be shown (and explained) to the instructor at the practicals in order to get 4% that count towards the overall mark.

1. (*) **Geometric Model for Populations.** Consider the geometric population model $P_{n+1} = (1+r)P_n$ for $n = 0, 1, \dots$ with solution $P_n = P_0 e^{sn}$ for $s = \log(1+r)$ as a model of Sweden's population from 1750 to 1960 and the US population from 1790 to 1990. The data for this are given in the worksheet `discrete.mw`. Explain what you see.
2. (*) **The Non-Linear Verhulst Model.** The general Verhulst model of population growth is as follows

$$P_{n+1} = \frac{1+r}{1 + \frac{rP_n}{K}} P_n$$

where $P_n \rightarrow K$ as $n \rightarrow \infty$. For the choice of parameters $P_0 = 2.79$, $r = 0.35$ and $K = 300$ compare the Verhulst model to the US population from 1790 to 1990. Explain what you see.

3. (*) **Population Model with Age Distribution.** Consider the human population divided into three age groups $0 - 14$, $15 - 39$ and ≤ 40 with population size P_i , $i = 1, 2, 3$ and yearly birth rates of 0, 0.06 and 0 and death rates of 0.005, 0.01 and 0.015, respectively.

(a) Show that $\mathbf{P}(t+1) = \mathbf{A}\mathbf{P}(t)$ where

$$\mathbf{P} = \begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 0.92867 & 0.06 & 0 \\ 0.0663 & 0.9504 & 0 \\ 0 & 0.0396 & 0.985 \end{bmatrix}$$

- (b) Find the largest eigenvalue of $\mathbf{A}$ and describe the long-term behaviour of this system.
- (c) Consider the growth of the population with the following initial populations. How long does it take for the total population to double in each case?
 - i. $P_1 = 200$, $P_2 = P_3 = 400$.
 - ii. $P_1 = 400$, $P_2 = P_3 = 300$.
 - iii. $P_1 = 200$, $P_2 = 500$ and $P_3 = 300$.

Explain what you see.

4. (*) **Chaotic Models.** Consider the Ricker model of Salmon population size as follows

$$P_{n+1} = re^{-\frac{P_n}{K}} P_n$$

where r is the rate and K is another parameter. For simplicity choose $K = 500$ and consider the behaviour of this system for

- (a) $r < 1$, with initial population of 1000.
- (b) $1 < r < e^2$, for initial population 100, 500 and 1000.
- (c) $r > e^2$, for initial population 100, 500 and 1000.

Explain what you see.